

Cornelia de Lange Syndrome 5 (CdLS5, HDAC8-related) — Comprehensive Disease Report

MONDO:0010471 · OMIM #300882 · Orphanet ORPHA:199 (CdLS group) · Gene: HDAC8 (Xq13.1)

Evidence source key: [H] human clinical/genetic; [M] model organism; [V] in vitro/biochemical; [C] computational/epigenomic. PMIDs cited throughout.

1. Disease Information

Overview. Cornelia de Lange syndrome type 5 (CdLS5) is the X-linked form of Cornelia de Lange syndrome caused by loss-of-function variants in *HDAC8*, the gene encoding histone deacetylase 8, the enzyme that deacetylates the cohesin subunit SMC3. CdLS is a multisystem congenital malformation and neurodevelopmental disorder characterized by distinctive facial features, growth restriction (pre- and postnatal), intellectual disability, upper-limb anomalies, hirsutism/hypertrichosis, and involvement of the cardiac, gastrointestinal, genitourinary and other systems [H,

[P 29995837](#) 30614194]. CdLS is a "cohesinopathy" and, more broadly, a **disorder of transcriptional regulation** (DTR) [H,

[P 37377026](#)]. HDAC8-related CdLS is frequently **non-classic/atypical**, and in many patients the clinical diagnosis was not suspected before genomic testing [H,

[P 24403048](#)].

Key identifiers. - MONDO: **MONDO:0010471** (Cornelia de Lange syndrome 5) - OMIM phenotype: **#300882**; OMIM gene *HDAC8*: **300269** - Orphanet: **ORPHA:199** (Cornelia de Lange syndrome, umbrella) - Gene: **HDAC8**, HGNC:13315, NCBI Gene 55869, Ensembl

ENSG00000147099, UniProt Q9BY41; cytoband **Xq13.1** - ICD-10: **Q87.1** (congenital malformation syndromes predominantly associated with short stature); ICD-11: **LD2F.1Y**/typically coded under multiple-anomaly syndromes; MeSH: **D003635** (De Lange Syndrome)

Synonyms/alternative names. CdLS5; HDAC8-related Cornelia de Lange syndrome; X-linked Cornelia de Lange syndrome; Cornelia de Lange syndrome due to HDAC8 deficiency; historically overlapping terms: Brachmann–de Lange syndrome. "Wilson–Turner-like" X-linked ID has been noted for some HDAC8 variants in the differential.

Data provenance. This report is compiled from **aggregated disease-level resources** (OMIM, Orphanet, HPO, consensus statements) and **published patient cohorts** (e.g., 38-patient HDAC8 cohort [PMID 24403048]; 716-proband CdLS cohort [PMID 37377026]); not from individual EHR data.

2. Etiology

Primary cause — genetic. Hemizygous (males) or heterozygous (females) loss-of-function variants in *HDAC8* [H,

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22885700]. HDAC8 is the vertebrate SMC3 deacetylase; CdLS-causing variants abolish enzymatic activity ("all cause a loss of enzymatic function") [H/V,

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Genetic risk factors. - *Causal variants:* predominantly **missense** and largely **de novo**; also nonsense, frameshift, splice, and deletions [H,

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- *Modifier of expressivity in females:* pattern of **X-chromosome inactivation (XCI)**. Skewed XCI silencing the mutant allele attenuates phenotype; random/unfavorable XCI increases severity [H,

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26671848]. - *Broader locus heterogeneity:* other CdLS genes (NIPBL, SMC1A, SMC3, RAD21, BRD4, ANKRD11) cause overlapping phenotypes; genetic background may modify.

Environmental risk factors. None established. CdLS is a monogenic Mendelian disorder; no confirmed toxic, infectious, dietary, or occupational cause. **Male sex** is a risk factor for greater severity in HDAC8-CdLS (hemizyosity, no XCI buffering) [H,

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Family history relevant when a carrier mother is mosaic.

Protective factors. The principal "protective" mechanism is **favorable (skewed) X-inactivation** in heterozygous females, which can render carriers mildly affected or clinically unaffected [H,

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No dietary/lifestyle protective factors are known. gnomAD constraint (HDAC8 highly intolerant to LoF) argues against benign LoF variation.

Gene–environment interactions. Not established; disease is essentially fully genetically determined. The main "gene–gene/epigenetic" interaction is XCI × mutant allele.

3. Phenotypes

HDAC8-CdLS overlaps classic CdLS but is often milder and has discriminating features. Frequencies below are drawn from CdLS-spectrum and HDAC8 cohorts [PMID 24403048; 30614194; 29995837]; HDAC8-specific frequencies are smaller-N.

Craniofacial (near-universal; congenital). - Synophrys / arched eyebrows — HP:0000664 / HP:0000574 (characteristic across CdLS) [PMID 30614194] - Short nose with anteverted nares, depressed bridge — HP:0003196 / HP:0000463 [PMID 30614194] - Long philtrum, thin upper vermilion — HP:0000343 / HP:0000219 [PMID 30614194] - Micrognathia — HP:0000347; low-set/posteriorly rotated ears - **HDAC8-discriminating:** delayed anterior fontanelle closure (HP:0001476), **ocular hypertelorism (HP:0000316)**, hooding of eyelids, broader nose, **dental anomalies (HP:0000164)** [H,

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"often displays delayed anterior fontanelle closure, ocular hypertelorism, hooding of the eyelids, a broader nose and dental anomalies, which may be useful discriminating features" [PMID 24403048].

Growth (congenital/childhood, chronic). Prenatal and postnatal growth restriction — HP:0001511 / HP:0001510; microcephaly (often postnatal) — HP:0000252 [PMID 30614194]. Severity generally milder in HDAC8 than NIPBL.

Neurodevelopment (childhood, lifelong). Intellectual disability — HP:0001249; global developmental delay — HP:0001263; speech delay — HP:0000750. HDAC8 cases trend milder cognitively than NIPBL but ID is typical.

Neurobehavioral/psychiatric (childhood → adult; persistent). - Autism spectrum features — HP:0000717 (significantly elevated vs comparison syndromes) [H,];

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ASD criteria may increase over time [PMID 30941551] - Anxiety — HP:0000739 (common, persistent; intolerance of uncertainty a driver) [PMID 36199025; 40878447] - Self-injurious behavior — HP:0100716; stereotypies — HP:0000733 - Catatonia-like attenuated behavior in ~30.3% of CdLS [PMID 29536582]

Limb (congenital). Upper-limb reduction defects (oligodactyly, absent forearm) — HP:0009821/HP:0002984 are **characteristic of severe NIPBL CdLS but typically ABSENT or mild in HDAC8-CdLS** [PMID 24403048; 30158690]. Small hands/feet, 5th-finger clinodactyly (HP:0004209) may occur.

Other systems. Hirsutism/hypertrichosis — HP:0001007; congenital heart defects — HP:0001627 (e.g., septal defects, pulmonary stenosis); gastro-esophageal reflux and GI dysmotility — HP:0002020; cryptorchidism/genital anomalies — HP:0000028; hearing loss — HP:0000365; ptosis — HP:0000508; myopia/ophthalmologic issues; seizures — HP:0001250 (subset). Male hemizygotes more severely affected [PMID 24403048].

Quality-of-life impact. Substantial: lifelong intellectual disability, communication limits, anxiety, self-injury, GI symptoms, and feeding difficulty impair adaptive functioning and caregiver burden [PMID 36199025; 29536582]. Disease-specific/formal QoL instrument data (EQ-5D/SF-36) are limited for CdLS5 specifically — **not available at HDAC8 subtype resolution.**

4. Genetic / Molecular Information

Causal gene. *HDAC8* (HGNC:13315; OMIM 300269), Xq13.1, encoding a class I zinc-dependent histone/lysine deacetylase; physiological substrate for the CdLS mechanism is **cohesin subunit SMC3** (acetyl-K105/K106) [H/V,

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Pathogenic variants. - *Types:* predominantly **missense** (clustered around catalytic residues/active site), plus nonsense, frameshift, splice-site, and intragenic/whole-gene deletions [H,

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Recurrent examples reported include p.His180Arg, p.Gly304 region, p.Thr311Met, and catalytic-domain substitutions (variant-level detail curated in ClinVar/HGMD). - *Classification (ACMG/AMP):* pathogenic/likely pathogenic for LoF and functionally validated missense; some VUS resolved by enzymatic assay or DNA-methylation epismutation. - *Allele frequency:* essentially **absent from population databases** (gnomAD) — consistent with a highly constrained, disease-causing gene. - *Origin:* **germline**, mostly **de novo**; **maternal mosaicism** documented (unaffected mother mosaic → two affected sibs) [H,

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HDAC8 somatic mutations occur in cancers but are unrelated to CdLS. - *Functional consequence:* **loss of function / loss of enzymatic (deacetylase) activity** [H/V,

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Modifier genes / factors. X-inactivation pattern is the dominant modifier in females [PMID 26671848]. NIPBL expression level correlates with severity across CdLS broadly [PMID 27125329]. No specific trans-modifier gene proven for *HDAC8*.

Epigenetic information. CdLS displays a reproducible genome-wide **DNA-methylation "epismutation"** usable diagnostically; it can confirm diagnosis in mutation-negative patients and reclassify VUS, though sensitivity is heterogeneous across CdLS cases [C/H,

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37872275]. Mechanistically, *HDAC8* loss alters SMC3 acetylation and cohesin-dependent chromatin architecture/transcription [V,

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Chromosomal abnormalities. CdLS5 is a single-gene disorder; large Xq13 deletions encompassing *HDAC8* can cause it and are detectable by CMA. Otherwise no characteristic karyotypic change.

5. Environmental Information

Environmental factors: none established. **Lifestyle factors:** none causal (monogenic disorder). **Infectious agents:** not applicable. (Environmental contribution is essentially nil; disease is genetically determined.)

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A loss-of-function *HDAC8* variant **results in** loss of HDAC8 lysine-deacetylase activity [H/V,

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2. Loss of HDAC8 activity **leads to** failure to deacetylate SMC3 (acetyl-K105/K106) during mitotic exit → **increased/retained SMC3 acetylation** [V,

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3. Retained-acetyl SMC3 **results in** inefficient dissolution of the "used" cohesin complex released from chromatin (prophase and anaphase) and its improper recycling/reloading [V,

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4. This **leads to** decreased occupancy at cohesin binding sites genome-wide → **altered chromatin architecture and gene transcription** (a pattern shared with NIPBL-mutant CdLS) [V,

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5. Dysregulated developmental transcription **results in** abnormal patterning/differentiation of multiple lineages, notably **cranial neural crest** — (*branch, demonstrated in mouse*): HDAC8 loss de-represses homeobox factors **Otx2 and Lhx1** in cranial neural crest, **leading**

to skull/craniofacial malformation [M,

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6. Multilineage transcriptional dysregulation during embryogenesis **results in** the CdLS phenotype: craniofacial dysmorphism, growth restriction, CNS/neurodevelopmental and behavioral abnormalities, limb, cardiac, and GI anomalies [H,

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7. (*Female branch*) Skewed X-inactivation silencing the mutant *HDAC8* allele **attenuates** the phenotype; hemizygous males lack this buffering and are **more severely affected** [H,

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26671848].

Note on inference: Steps 1–4 are biochemically demonstrated in human cells; step 5 (Otx2/Lhx1) is demonstrated in mouse neural crest and inferred to underlie human craniofacial features; step 6 is the clinical–molecular correlation.

Detail by category

- **Molecular pathways:** cohesin cycle / sister-chromatid cohesion establishment–dissolution; cohesin-mediated chromatin looping and enhancer–promoter regulation (Reactome "Cohesin Loading onto Chromatin," "Establishment of Sister Chromatid Cohesion"). Downstream developmental TF networks (Otx2, Lhx1). Not a classic signaling cascade (Wnt/MAPK) primarily, though cohesin regulates many such loci.
- **Cellular processes:** GO:0016575 histone deacetylation; GO:0007062 sister chromatid cohesion; GO:0006325 chromatin organization; GO:0006351 transcription; cell-fate specification/patterning; cranial neural crest migration/differentiation. Increased apoptosis of neural crest in *Hdac8*-null skull [M,

P 19605684

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- **Protein dysfunction:** loss of catalytic deacetylase activity (Zn²⁺-dependent hydrolase, InterPro/Pfam PF00850 Hist_deacetyl; PDB structures of HDAC8 available). Missense variants disrupt catalysis/substrate binding → LoF [V,

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- **Biochemical abnormality:** enzyme deficiency — SMC3-lysine deacetylase (HDAC8) [V,

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- **Epigenetic changes:** aberrant SMC3 acetylation; altered histone deacetylation of neural-crest TF loci; reproducible blood DNA-methylation epismutation [PMID 22885700; 38751117].
- **Immune/metabolic involvement:** not central; no primary immunodeficiency/metabolic defect characteristic of CdLS5.
- **Transcriptomics:** CdLS cell lines (NIPBL or HDAC8 mutant) share a consistent altered transcription profile with reduced cohesin site occupancy [V,

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- **Cell types (CL) / anatomy (UBERON):** CL:0000333 cranial neural crest cell; UBERON:0003129 skull; UBERON:0001890 forebrain.

7. Anatomical Structures Affected

- **Organ/system level (primary):** craniofacial skeleton/skull (UBERON:0003129), central nervous system/brain (UBERON:0000955), musculoskeletal/limbs (upper limb UBERON:0002102). **Secondary/associated:** heart (UBERON:0000948), gastrointestinal tract incl. esophagus (UBERON:0001043)/gut, genitourinary tract (UBERON:0000079), ears/auditory system (UBERON:0001690), eyes (UBERON:0000970), skin/hair (hypertrichosis).
- **Body systems:** nervous, craniofacial/skeletal, cardiovascular, digestive, genitourinary, integumentary, sensory (auditory/visual), endocrine (growth).
- **Tissue/cell level:** connective/skeletal (cranial bones), nervous tissue; key developmental population = **cranial neural crest cells (CL:0000333)** [M,

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- **Subcellular (GO cellular component):** GO:0005634 nucleus; GO:0000785 chromatin; GO:0008278 cohesin complex; GO:0005737 cytoplasm (HDAC8 also cytoplasmic).
- **Localization/lateralization:** malformations are generally **bilateral/symmetric** (craniofacial midline and paired structures); limb involvement, when present in the broader spectrum, can be asymmetric.

8. Temporal Development

- **Onset: congenital** — malformations arise in embryogenesis; growth restriction detectable prenatally; dysmorphism recognizable at birth [H, [P 29995837](#)].
Onset pattern: chronic/static malformation with developmental unfolding.
- **Progression:** Non-progressive at the malformation level; **lifelong/chronic** course. Neurobehavioral features (autism traits, anxiety, catatonia-like attenuation) can **emerge or intensify across childhood into adulthood** [H, [P 30941551](#)];
29536582]. Adaptive ability tends to be stable with some receptive-language gains [PMID 30941551].
- **Patterns:** No remission (structural/genetic). **Critical period:** early embryonic neural-crest/organogenesis window sets the malformation; postnatally, early developmental/behavioral intervention is the actionable window.

9. Inheritance and Population

- **Epidemiology:** CdLS (all genes) estimated incidence/prevalence ~**1/10,000–1/30,000** [H, [P 32800026](#)].
HDAC8 accounts for a **minority (~4–5%)** of CdLS; CdLS5 is correspondingly rarer. Precise CdLS5 prevalence: **not separately established**.
- **Inheritance: X-linked** (Xq13.1). De novo in most; can be inherited from a carrier/mosaic mother [PMID 24403048; 26671848].
- **Penetrance/expressivity:** Highly **variable expressivity**; in females penetrance/severity is modulated by **X-inactivation** (can be near-nonpenetrant with fully skewed favorable XCI) [PMID 26671848]. Males: essentially penetrant, more severe.
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Germline/somatic mosaicism:** documented **maternal mosaicism** with sibling recurrence [PMID 26671848]; relevant to recurrence-risk counseling.

- **Founder effects/consanguinity/carrier frequency:** no founder effect; consanguinity not relevant (X-linked, mostly de novo); carrier frequency negligible in general population (gnomAD LoF depleted).
- **Population demographics:** No ethnic predilection; CdLS occurs across global populations with consistent core features [H,

P 30614194

Sex ratio: both sexes affected; **males more severely affected**, females often milder/atypical [PMID 24403048]. Age distribution: present from birth through adulthood.

10. Diagnostics

- **Clinical criteria: 2018 first international consensus** provides diagnostic criteria for **classic and non-classic CdLS** and a cardinal/suggestive feature scoring approach; recommends molecular confirmation [H,

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Facial-analysis technology aids recognition across populations (sensitivity $\geq 95\%$, specificity $\geq 91\%$) [PMID 30614194].

- **Genetic testing (recommended, definitive):**
- **Multigene NGS panel / WES / WGS** covering the seven CdLS/DTR genes incl. *HDAC8* — first-line; HDAC8 sequencing is "indispensable" in CdLS workup [H,

P 26671848

WGS adds detection of structural/deep-intronic variants.

- **Single-gene *HDAC8* sequencing** when HDAC8 phenotype suspected.
- **Chromosomal microarray (CMA)** to detect Xq13 deletions/CNVs.
- **Deep/mosaicism-aware sequencing of multiple tissues** (buccal, fibroblasts) because somatic/parental mosaicism is common in CdLS [PMID 26671848].
- **X-inactivation assay** in females to interpret expressivity [PMID 24403048].
- **Omics-based diagnostics: DNA-methylation epsignature** as an orthogonal classifier to confirm mutation-negative CdLS and reclassify VUS [C,

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Functional **SMC3-deacetylation/enzyme assays** can validate missense VUS [V,

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- **Supportive clinical tests:** echocardiography (CHD), hearing and ophthalmologic evaluation, GI/reflux workup, renal ultrasound, growth monitoring, developmental/behavioral assessment (per consensus surveillance) [PMID 29995837].
- **Differential diagnosis:** classic NIPBL-CdLS and other cohesinopathy genes (SMC1A, SMC3, RAD21); CdLS-like DTRs (ANKRD11/KBG, EP300, AFF4/CHOPS, TAF1, BRD4); Warsaw breakage syndrome (DDX11), Roberts syndrome (ESCO2), fetal alcohol spectrum, Coffin–Siris, Rubinstein–Taybi [H,

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36703504].

- **Screening:** No population newborn screening (not applicable). **Cascade/carrier and prenatal testing** offered once a familial variant is known; prenatal ultrasound may show IUGR, limb/heart anomalies.

11. Outcome / Prognosis

- **Survival/life expectancy:** Many individuals survive into adulthood, particularly milder (incl. many HDAC8) cases; life expectancy is influenced by complications (severe CHD, GI/aspiration, infections). Formal CdLS5-specific survival statistics: **not established**. Hemizygous males (more severe) have poorer prognosis than mildly affected skewed-XCI females [PMID 24403048].
- **Morbidity/function:** Lifelong intellectual disability and behavioral morbidity (anxiety, ASD, self-injury, catatonia-like features) dominate disability burden [PMID 36199025; 29536582]. Growth failure, feeding/GI problems, hearing/vision deficits contribute.
- **Complications:** GERD and GI dysmotility, aspiration, recurrent infections, congenital heart disease, seizures, behavioral crises/self-injury.
- **Recovery potential:** Malformations are static; developmental and behavioral outcomes are improvable with early intervention/therapy; no cure.

- **Prognostic factors:** genotype/gene (HDAC8 milder than NIPBL), **sex and X-inactivation pattern**, severity of CHD/GI disease, degree of ID. No validated molecular prognostic biomarker beyond gene/variant and XCI [PMID 24403048; 27125329].

12. Treatment

No disease-modifying/curative therapy exists; management is **multidisciplinary and supportive**, per the 2018 consensus [H,].

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- **Supportive/medical (NCIT: C15277 Supportive Care):**
- **GI:** anti-reflux medical therapy (proton-pump inhibitors — NCIT C29708; H2 blockers), nutritional support, fundoplication/gastrostomy for severe GERD/feeding failure.
- **Cardiac:** surgical/medical management of congenital heart defects.
- **ENT/audiology:** hearing aids, myringotomy tubes for otitis; ophthalmology for ptosis/refractive error.
- **Growth/endocrine:** nutritional optimization; growth monitoring.
- **Neurology:** anti-seizure medication if epilepsy.
- **Neurobehavioral/psychiatric (NCIT C15313 Psychosocial/Behavioral therapy; C265 Pharmacotherapy):** behavioral therapy for self-injury/ASD; SSRIs (e.g., sertraline — NCIT C47727) for anxiety/OCD-like symptoms; treat pain/occult GI sources of behavioral change; address intolerance of uncertainty in anxiety interventions [PMID 40878447; 36199025].
- **Surgical/interventional (NCIT C15329 Surgery):** cardiac repair, fundoplication/gastrostomy, orchidopexy for cryptorchidism, cleft/orthodontic and other reconstructive procedures as indicated.
- **Rehabilitative:** physical, occupational, and speech/language therapy; special education; early developmental intervention.
- **Advanced/experimental therapeutics: None approved.** No gene, cell, RNA, or targeted therapy in clinical use for CdLS5. HDAC-modulation strategies are conceptual/preclinical only; no CdLS-specific trials identified. Pharmacogenomics: no CdLS5-specific PGx guidance.
- **Treatment strategy:** individualized, guided by the consensus care/surveillance pathway; genotype-informed counseling but not yet genotype-targeted drug therapy [PMID 29995837].

13. Prevention

- **Primary prevention:** Not possible (mostly de novo genetic disorder). Prevention centers on **reproductive counseling** for families with a known variant.
- **Secondary prevention:** Early molecular diagnosis → structured **surveillance** (cardiac, GI, hearing, vision, growth, behavioral) to detect and treat complications early [PMID 29995837].
- **Tertiary prevention:** Aggressive management of GERD/aspiration, CHD, and behavioral/self-injury to prevent morbidity.
- **Genetic screening/counseling: genetic counseling** (X-linked risks, de novo vs mosaic recurrence), **prenatal testing / PGT** when a familial variant is known; test/counsel potentially mosaic mothers given documented sibling recurrence [PMID 26671848]. **NSGC/ACMG** counseling frameworks apply.
- **Immunization/public-health/environmental measures:** not applicable beyond routine pediatric care.

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *HDAC8* is conserved across vertebrates. Mouse *Hdac8* (NCBI Gene 70315; NCBI Taxon 10090); also conserved in rat, zebrafish. Cohesin/SMC3 machinery conserved from yeast (Hos1 is the yeast SMC3 deacetylase analog) to humans [PMID 22885700].
- **Natural disease in other species:** No well-characterized naturally occurring HDAC8-CdLS in companion/wild animals reported in OMIA; **not applicable/none documented**. Veterinary relevance: none established.
- **Comparative biology:** The cohesin acetylation cycle and neural-crest role of *Hdac8* are evolutionarily conserved, underpinning cross-species disease modeling [PMID 22885700; 19605684].
- **Transmission/zoonosis:** not applicable (genetic disorder).

15. Model Organisms

- **Mouse (*Mus musculus*, NCBI Taxon 10090) — primary model [M,**

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- **Global *Hdac8* knockout:** perinatal lethality due to **skull instability** — models the craniofacial malformation.
- **Conditional (cranial neural crest–specific, e.g., *Wnt1-Cre*) knockout:** **phenocopies** the skull defect, localizing pathogenesis to cranial neural crest.
- **Mechanistic readout:** HDAC8 represses homeobox TFs **Otx2 and Lhx1**; loss de-represses them → aberrant skull patterning; increased neural-crest apoptosis.
- **Recapitulation:** strong for craniofacial/skull patterning; **limitations:** global KO perinatal lethality limits study of postnatal neurobehavioral phenotypes; does not capture X-inactivation mosaicism of human females.
- **Zebrafish (*Danio rerio*) / cohesinopathy models:** cohesin (*nipbl*, *esco2*, *rad21*) models reproduce craniofacial and developmental gene-expression phenotypes, supporting the transcriptional-dysregulation mechanism shared with HDAC8 [M,

P 29084713

- **In vitro / cellular models:** patient-derived lymphoblastoid/fibroblast lines showing increased SMC3 acetylation, reduced cohesin occupancy, and altered transcription [V,

P 22885700

recombinant HDAC8 enzymatic assays to test variant activity [V,

P 24403048

iPSC/organoid HDAC8-CdLS models: an emerging but not yet standard resource.

- **Model databases:** MGI (mouse *Hdac8*), IMPC, ZFIN (zebrafish), Alliance of Genome Resources.
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Summary Answer

Cornelia de Lange syndrome 5 is the **X-linked (Xq13.1) form of CdLS caused by loss-of-function variants in *HDAC8***, the deacetylase for the cohesin subunit SMC3. Impaired SMC3 deacetylation disrupts the cohesin acetylation/recycling cycle and cohesin-dependent transcriptional regulation during development (a "cohesinopathy"/disorder of transcriptional regulation), producing a multisystem malformation and neurodevelopmental syndrome that overlaps classic CdLS but is often **milder/atypical**, with discriminating features (delayed fontanelle closure, ocular hypertelorism, dental anomalies); **hemizygous males are more severely affected**, while heterozygous females are variably affected depending on X-inactivation. Management is multidisciplinary and supportive with no disease-modifying therapy; diagnosis rests on consensus clinical criteria plus molecular testing (with episignature/functional assays as adjuncts).

Key Limitations

- Many phenotype frequencies and prognosis/QoL data are **CdLS-spectrum-level**, not HDAC8-specific (small N for CdLS5).
- No CdLS5-specific survival, incidence, or clinical-trial data; therapeutics remain supportive.
- Variant-level ClinVar/gnomAD counts were not directly queried in this literature-based synthesis and should be verified in those databases for KB population.

Primary References (PMIDs)

22885700 (Deardorff 2012, *Nature*, HDAC8=SMC3 deacetylase/mechanism); 24403048 (Kaiser 2014, 38-patient HDAC8 cohort, X-linked, discriminating features); 26671848 (Parenti 2016, HDAC8 spectrum, skewed XCI, maternal mosaicism); 19605684 (Haberland 2009, *Hdac8* mouse skull/neural crest, Otx2/Lhx1); 29995837 (Kline 2018, first international consensus); 37377026 (Kaur 2023, 716 probands, DTR framing); 30614194 (Dowsett 2019, diverse populations); 30158690 (Yuan 2019, mild end of spectrum); 27125329 (Kaur 2016, NIPBL levels/severity); 23937369, 36199025, 29536582, 30941551, 40878447 (CdLS neurobehavioral phenotype); 38751117, 37872275 (episignature diagnostics); 31721174 (chromatinopathies review); 29084713 (zebrafish cohesinopathy model).

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